

M2_3 Extra: Experimental features of Bistability

1. Apparently contradictory findings under the same experimental conditions.

The unsuspected presence of bistability might lead to the erroneous conclusion that the preparation of experimental conditions was not done properly because the stable states differ quantitatively (expression level of a gene) even qualitatively (e.g. different phenotypes associated with different expression levels). Experimentally, a transition from a monostable to a bistable gene expression profile was found during the progression of a bacterial infection model of cholera (Nielsen et al. 2010). The result was that originally identical cells could be induced to evolve into one of two different virulent gene expression profiles.

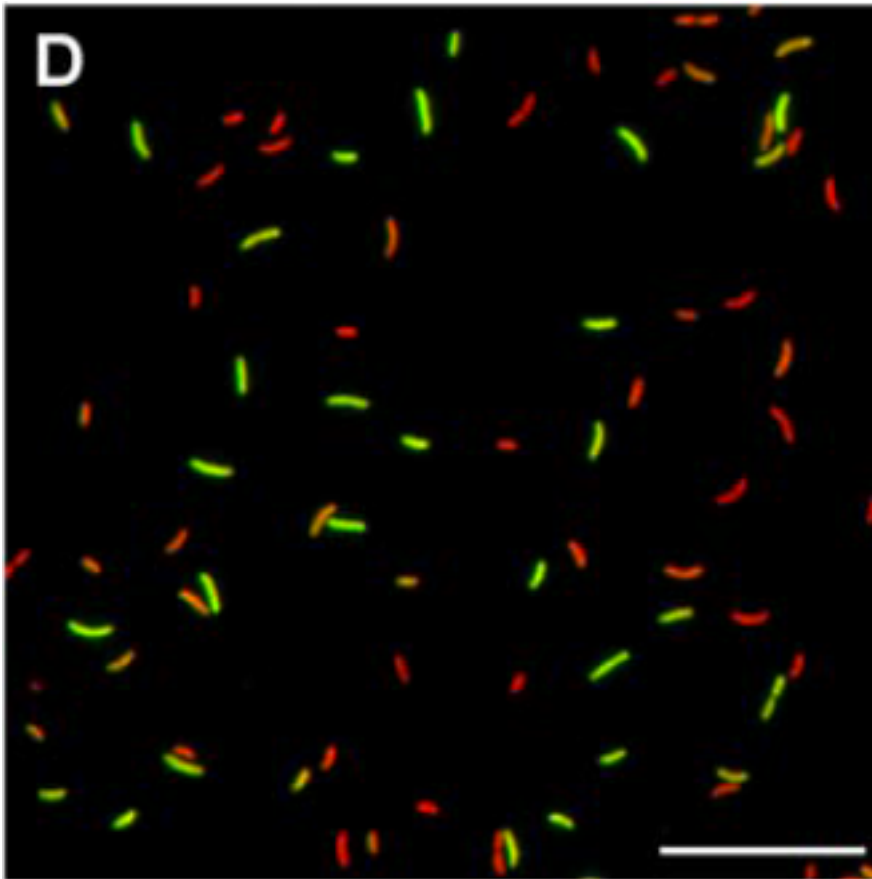
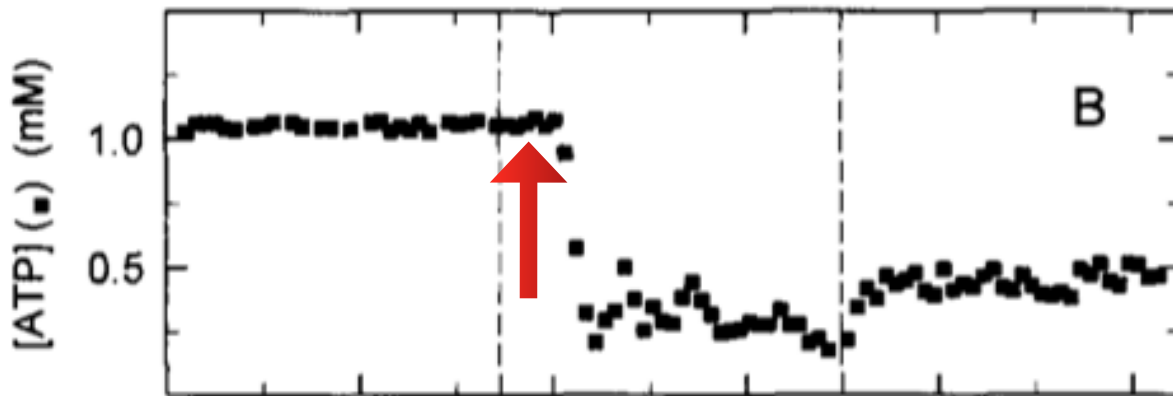


Figure 5D from Nielsen et al (2010) of bacteria with different expression levels.

In metabolic systems two different fluxes might be observed under same open-flow conditions. An example of the latter is given in Frenzel et al. (1996) who studied glycolysis in cell-free extracts from rat liver. They observed two different ATP levels under they same open flow conditions (representing constant supply of nutrients).

Figure 1B from Frenzel et al. (1996). Horizontal axis is time (total 1000 minutes). Experimental conditions in the left and right third of the experiment are under identical



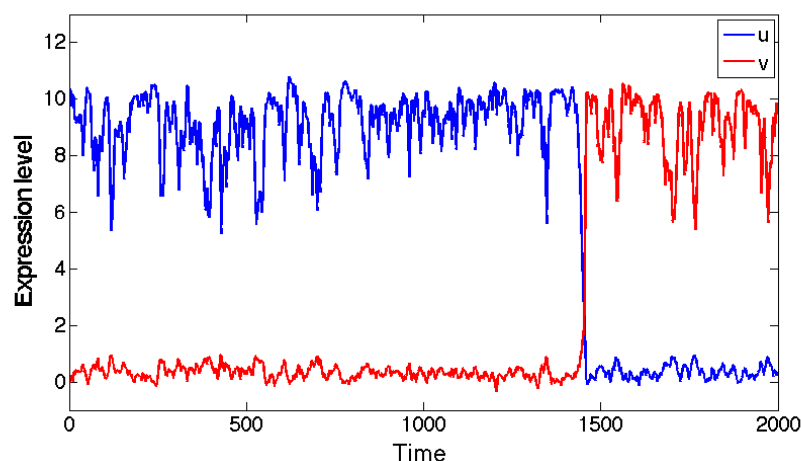
conditions. Central third represents the switching from the high ATP state to the low ATP state.

2. Single pulse can induce a switch

If bistability is suspected, it is possible to test for it. Notably, short perturbations are able to induce a permanent switch. In the experiment by Frenzel et al., the transition could be induced by transient addition of glycolysis regulator Fructose-2,6-bisphosphate (arrow). It is important that only pulses that push the system over the separatrix between the two states are successful in switching. Thus, knowledge of the state space, e.g. via a model equation, serves to predict the appropriate type of perturbation.

3. Noise can induce spontaneous switches

The presence of noise or noisy fluctuations in a bistable situation may lead to spontaneous transitions from one state to the other. This happens if the noise-induced fluctuations of the stable state are strong enough to cross the separatrix. The figure shows the simulation of the toggle switch model with independent noise terms added to both variables.



4. Bistability distributions can be bimodal

If under noisy conditions the two states in a bistable system coexist, the distribution of state-related features will be bimodal instead of unimodal. In gene expression experiment by Nielsen et al. fluctuations during the transition from the monostable to the bistable state lead to a bimodal bacterial population distribution.

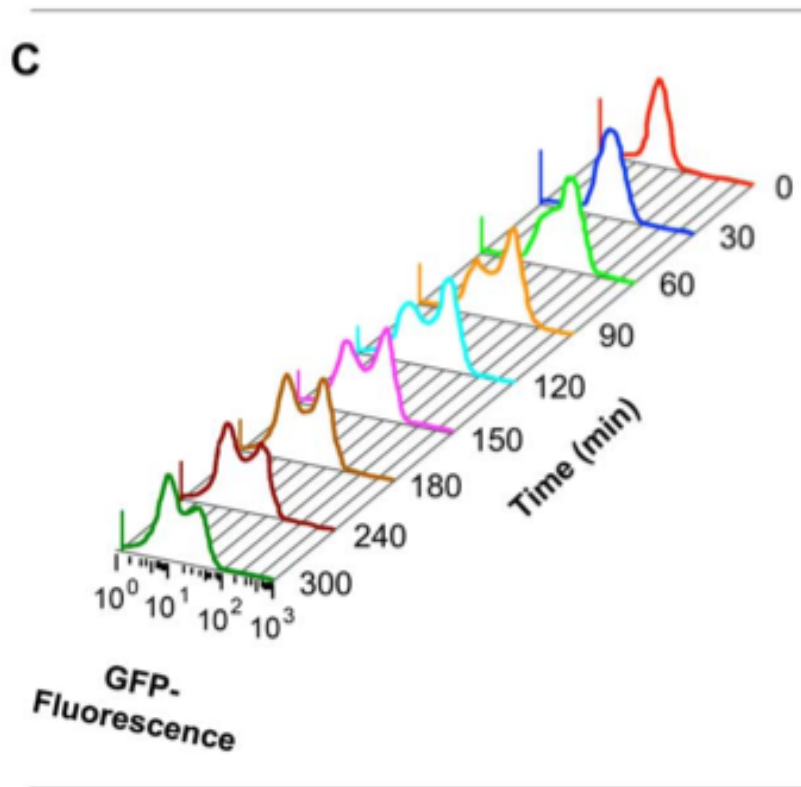
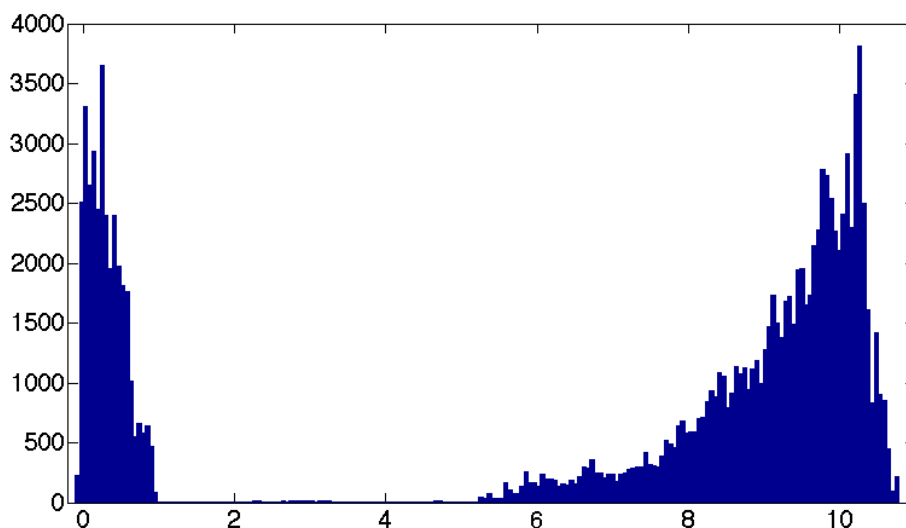


Figure 6C from Nielsen et al (2010) of the transition from monostable (Time=0) to bistable (Time >60minutes).

Here is a bimodal histogram of a long time series from the noisy toggle switch in section 3.



5. Bistability in Space

An interesting extension of the concept of bistability is found in a multicellular context. There it is possible to have e.g. different gene expression levels not erratically mixed as in the case of bacterial populations above. Rather, the two different levels might be properly organised in space. An example was given by Lopes et al. (2008) in the *Drosophila* embryo, where key development genes self-organise into spatial patterns of up and down state.

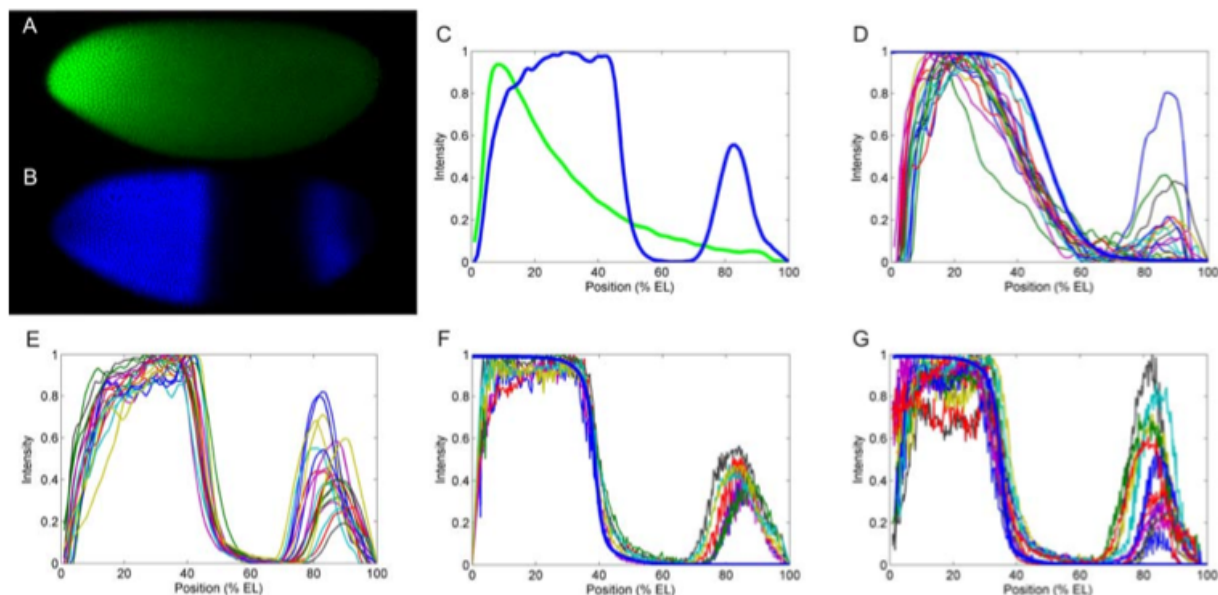


Figure 1 A-G from Lopes et al (20089). Gene expression level is imaged and plotted as a function of location along the embryo (anterior left; posterior right).

6. Bistability in Disease

Bistability is a self-organised feature of nonlinear systems. It is observed at different levels of organisation. A major application could be the role of bistability in cell fate decisions during differentiation and maturation. Xiong and Ferrell (2003) have highlighted this experimentally for the maturation of *Xenopus* oocytes. They showed that cell fate decision can be performed using transient stimuli in a bistable setting of gene expression.

Based on modelling work researchers have suggested certain brain diseases to present examples of bistability. The idea is that one of the two states corresponds to a normal state and the other to a diseased state. For example, certain types of epileptic seizures were claimed to be examples of noise-induced transitions between a normal and a pathological state of brain activity (Lopes da Silva et al. 2003). The spreading depression of migraine could form a sudden transition to a bistable abnormal state which later on self-resets (Dahlem and Isele, 2013). And finally, the special case of two pathological states with spontaneous switching between them could be underlying bipolar disorders. In the previous cases, a potential dynamical treatment would consist of applying a proper perturbation to reach the normal state. However, in the case of a bipolar disorder, one would rather like to avoid either state and (at least according to the model) stabilise the unstable central state (see e.g. Wehr et al. 1998).

7. References

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